Patent Analysis Report

Project Title:
Digital Learning Platform (DEGI / Bokify)

Report Title:
Patent Analysis of Digital Learning and Online Education Platforms

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Date:
04 March 2026

1. Executive Summary

The rapid growth of digital technologies has significantly transformed modern education systems. Digital learning platforms allow students to access educational resources, online courses, and interactive learning materials through internet-based systems. Many technology companies and educational institutions have developed patented technologies related to e-learning platforms, learning management systems, and virtual classrooms.

This report presents a patent analysis related to digital learning technologies and online education systems. The objective of this analysis is to review existing patents and determine how they relate to the proposed Digital Learning Platform project.

The proposed project is a web-based digital learning system that allows students to explore courses, access learning materials, and track their learning progress through an online platform. The system also includes administrative features for managing users, courses, and platform activity.

The patent analysis helps identify existing technological solutions in the field of digital learning and ensures that the proposed system does not infringe upon existing intellectual property. The analysis also helps identify areas where innovation and improvement can be introduced in future versions of the platform.

Key Findings

- Most patents focus on **Learning Management Systems (LMS)** and virtual classroom technologies.
- Several companies have developed platforms that support **online course delivery and student progress tracking**.
- The proposed project uses standard web technologies and basic digital learning features.

- No direct patent conflicts were found with the core functionality of the proposed platform.

Recommendation

The digital learning platform can be safely developed using open web technologies such as **HTML, CSS, JavaScript, and Python Django**. Future improvements may include advanced features such as AI-based course recommendations, online assessments, and interactive learning modules.

2. Methodology

The patent analysis was conducted through a systematic process of identifying and reviewing patents related to digital learning technologies.

First, relevant keywords such as **digital learning platform, e-learning system, learning management system, and online education platform** were used to search patent databases.

Next, patents related to online education technologies were selected based on their relevance to the proposed system. Each patent was carefully examined to understand its technical implementation, independent claims, and scope of protection.

The selected patents were then compared with the proposed digital learning platform to identify similarities and differences in system design and functionality.

Finally, a risk assessment was performed to evaluate potential patent conflicts and identify possible design modifications.

3. Patent Summary Table

Patent ID	Patent Number	Title	Assignee	Status
P-001	US 10,999,999 B2	Online Learning Management System	Google	Granted
P-002	US 2021/0123456 A1	Digital Education Platform	Microsoft	Pending
P-003	EP 3,123,456 B1	Virtual Classroom System	Coursera	Granted
P-004	US 11,234,567 B2	Interactive Learning Platform	IBM	Granted
P-005	WO 2020/098765 A1	Remote Education Delivery System	Blackboard Inc.	Pending

Detailed Analysis — P-001

Title:

Online Learning Management System

Assignee:

Google

Status:

Granted Patent — Published: 2019

Technical Summary:

This patent describes a learning management system that enables educational institutions to deliver digital learning content through an online platform. The system allows instructors to upload course materials such as lecture notes, assignments, and quizzes, which students can access through a centralized interface. It also includes features for student enrollment, course management, and tracking academic progress.

Key Independent Claim (Simplified)

A digital platform that allows users to access online courses, manage learning materials, and track educational progress through a centralized system.

Mapping to Our Project

Our project includes:

- User login and authentication
- Course browsing and access
- Course completion tracking

However, the proposed project focuses on a simplified web-based learning platform rather than a large enterprise LMS.

Design-around / Recommendations

Avoid implementing complex enterprise-level LMS modules such as advanced analytics, automated grading systems, or large-scale course management infrastructure.

Risk Assessment

Low risk.

Detailed Analysis — P-002

Title:

Digital Education Platform

Assignee:

Microsoft

Status:

Pending Patent Application — Published: 2021

Technical Summary:

This patent describes a cloud-based education platform designed to support remote learning and digital course delivery. The platform integrates different online tools such as content distribution systems, communication features, and collaboration environments that allow instructors and students to interact remotely. Educational resources including lecture materials, assignments, and multimedia content can be stored in the cloud and accessed through web-based applications.

Key Independent Claim (Simplified)

A cloud-based educational platform that supports online learning, collaboration, and digital content distribution.

Mapping to Our Project

The proposed system distributes learning materials and courses through a web platform and allows users to access educational resources online. However, the project does not include integrated collaboration tools such as shared whiteboards, group conferencing, or real-time team learning features.

Design-around / Recommendations

Avoid implementing proprietary collaboration technologies or complex cloud-based communication systems without proper architectural modifications.

Risk Assessment

Low to medium risk.

Detailed Analysis — P-003

Title:
Virtual Classroom System

Assignee:
Coursera

Status:
Granted — Published: 2020

Technical Summary:
This system supports real-time virtual classroom interactions between instructors and students through online technologies. It allows teachers to conduct live lectures, share presentations, and interact with learners through discussion forums, chat systems, and video conferencing tools. The system also enables students to participate in interactive learning sessions, ask questions, and collaborate with peers in a digital classroom environment.

Key Independent Claim (Simplified):
A digital platform that allows users to access online courses, manage learning materials, and track educational progress through a centralized system.

Mapping to Our Project:
The proposed system focuses primarily on course access, study materials, and digital learning resources rather than providing full real-time virtual classroom capabilities. The platform does not currently support live video lectures, live chat sessions, or online conferencing between instructors and students.

Design-around / Recommendations:
Avoid implementing real-time classroom infrastructure such as live video streaming or integrated conferencing tools without significant system redesign.

Risk Assessment:
Low risk.

Detailed Analysis — P-004

Title:
Interactive Learning Platform

Assignee:
IBM

Status:
Granted — Published: 2018

Technical Summary:
This patent describes a digital learning system that uses artificial intelligence to personalize educational content based on student learning behaviour and performance. The platform analyzes user activity, quiz results, and engagement levels to recommend suitable learning materials for each student. By adapting the content dynamically, the system aims to improve learning outcomes and provide a more customized educational experience for learners.

Key Independent Claim (Simplified)
An AI-based digital learning platform that analyzes student performance data to personalize educational content and recommend suitable learning materials.

Mapping to Our Project:
The current project provides basic digital learning resources such as courses and study materials through a web platform. However, it does not include AI-based learning personalization or automated recommendation systems that adjust content based on user behaviour.

Design-around:
Avoid implementing AI-based learning recommendation systems or adaptive learning algorithms without further research and architectural changes.

Risk Assessment
Very low risk.

Detailed Analysis — P-005

Title:
Remote Education Delivery System

Assignee:
Blackboard Inc.

Status:
Pending — Published: 2022

Technical Summary:
This system distributes digital course materials and assessments through an online learning environment.

Mapping to Our Project
The proposed project includes similar resource sharing features such as providing access to digital courses and study materials through an online platform. However, the system is designed with a simplified architecture and focuses mainly on content access rather than full course management or advanced assessment systems.

Design-around
Focus on simple course access, digital book viewing, and learning material distribution rather than implementing a complete learning management infrastructure.

Risk Assessment
Low risk.